

# Gao's theorem for nonabelian groups

OPG-337

L1 - TRACTABLE

OPEN

Number Theory

## ORIGINAL PROBLEM STATEMENT

For every finite multiplicative group  $G$ , let  $s(G)$  (  $s'(G)$  ) denote the smallest integer  $m$  so that every sequence of  $m$  elements of  $G$  has a subsequence of length  $> 0$  (length  $|G|$  ) which has product equal to 1 in some order.

Conjecture  $s'(G) = s(G) + |G| - 1$  for every finite group  $G$ .

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Archive: <https://www.unsignedmath.com/problems?difficulty=1>

Dataset: <https://huggingface.co/datasets/ulamai/UnsolvedMath>